MATLAB

matlab.mathworks.com

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Search (Ctrl+Shift+Space)

KARTHIK

Save As | Copy Figure | Print | Show Code | FILE

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

LABELS AND ANNOTATIONS

/ › MATLAB Drive ›

Files

Name

- 5G AND 6G LAB
- LAB EXPERIMENT
- New Folder
- ASSEMENT QNO 1.m
- EX : 2.m
- ex 1.m
- EX 2 ..m
- EX 2..m
- Ex 2.m
- ex 5.m
- ex1.m

Workspace

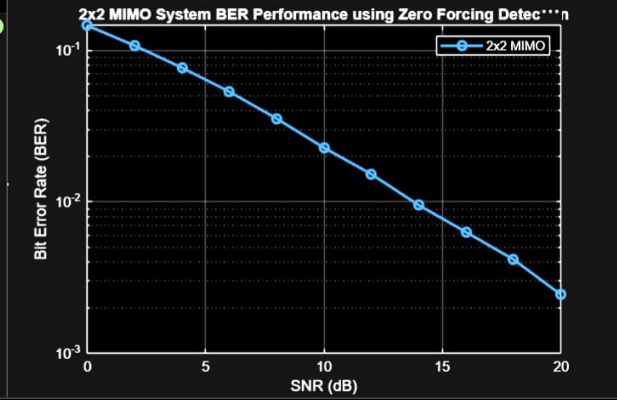
Name	Value	Size	Class
BER	1×11 double	1×11	double
bits	2×100000 d...	2×100000	double
detected	2×100000 d...	2×100000	double
H	2×2×10000...	2×2×100000	double (com...
i	100000	1×1	double
k	11	1×1	double
N	100000	1×1	double
noise	[-0.0044 - 0....	2×1	double (com...
noiseVar	0.0100	1×1	double
Nr	2	1×1	double

EX1.m | ex2.m | ex3.m | EX4.m | untitled4.m | EX6.m

/MATLAB Drive/untitled4.m

```
35          x_hat = pinv(H(:,:,i))*y;
36
37          % BPSK Decision
38          detected(:,i) = real(x_hat) > 0;
39
40      end
41
42      % BER Calculation
43      BER(k) = sum(bits(:) ~= detected(:))/numel(bits);
44
45  end
46
47  % BER Plot
48  figure;
49  semilogy(SNRdB,BER,'-o','LineWidth',2);
50  grid on;
51  xlabel('SNR (dB)');
52  ylabel('Bit Error Rate (BER)');
53  title('2x2 MIMO System BER Performance using Zero Forcing Det
54  legend('2x2 MIMO');
```

Figure 1



Command Window

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36°C Partly sunny

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